

Baye's Theorem: Let the events $A_1, A_2, A_3, \dots, A_n$ form a partition of the sample space, S , such that $P(A_j) > 0$ for $j=1, 2, 3, \dots, n$ and Let B be an event such that $P(B) > 0$.

Then,

A_1	A_2	A_3	A_4	A_5	A_n	B
$\frac{P(A_1 \cap B)}{P(B)}$						

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{\sum_{j=1}^n P(A_j) \cdot P(B | A_j)}$$

e.g.

$$P(A_1 | B) = \frac{P(A_1) \cdot P(B | A_1)}{\sum_{j=1}^n P(A_j) \cdot P(B | A_j)} \rightarrow \text{Total probability}$$

f: Using the defⁿ of Conditional Probability

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)} \quad \text{--- (1)}$$

Since the events $A_1, A_2, A_3, \dots, A_n$ form a partition in S , and B be any event in S . [B cannot be outside S]

$$B = S \cap B$$

$$S = (A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) \cap B$$

$$= (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B) \dots \cup (A_n \cap B)$$

[Using distributive Law]

Therefore,

$$P(B) = P[\overset{\text{mutually disjoint}}{\uparrow} (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B) \cup \dots \cup (A_n \cap B)]$$

$$\Rightarrow P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B) + \dots + P(A_n \cap B) \quad [\text{Since the events are mutually exclusive}]$$

$$\Rightarrow P(B) = \sum_{j=1}^n P(A_j \cap B) \quad \text{--- ②}$$

Now, using ② eqn, ① can be rewritten as

$$P(A_i | B) = \frac{P(A_i \cap B)}{\sum_{j=1}^n P(A_j \cap B)} \quad \text{--- ③}$$

According to the multiplication Law,

$$P(A_j \cap B) = P(A_j) \cdot P(B | A_j) \quad \text{--- ④}$$

Using ①, eqn ③ can now be written as

$$\left[P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{j=1}^n P(A_j) \cdot P(B|A_j)} \right] \text{ for } i=1,2,3,\dots,n$$

[Proved]

Prior Probabilities $\rightarrow$ Known

Posterior $\rightarrow$ (initial) unknown

Proof: (Identifying the source of defective items)

In a factory, three different machines M_1, M_2, M_3 were used for producing a large batch of similar manufactured items. Suppose that M_1, M_2, M_3 produced respectively 25%, 35%, and 40% of the total items. $\rightarrow$ Prior

Suppose further that of their produced items, respectively 5%, 4%, and 2% are defective. Finally, one item is selected at random from the entire batch

and it is found to be defective.
Determine the probability that the selected item was produced by M_1 , M_2 or M_3 .

Let us define the events as follows:

A_1 : The selected item was produced by M_1 .

A_2 : " " " " " " M_2 .

A_3 : " " " " " " M_3 .

B : The " " " defective.

B is such an event whose participation is in all A s.

Then we have,

$$P(A_1) = 25\% = 25/100 = 1/4 = 0.25$$

$$P(A_2) = 0.35, \quad P(A_3) = 0.40$$

$$P(B|A_1) = 5\% = 0.05$$

$$P(B|A_2) = 4\% = 0.04$$

$$P(B|A_3) = 2\% = 0.02$$

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}$$

$$= \frac{0.25 \times 0.05}{(0.25 \times 0.05) + (0.35 \times 0.04) + (0.40 \times 0.02)}$$

$$= 0.36$$

Approximately, 36% of the defective items have been produced by machine, M_1 .

b) $P((A_2 \cup A_3) | B) = P(A_2 | B) + P(A_3 | B)$
or (disjoint)

$$= \frac{0.35 \times 0.04}{0.0345} + \frac{0.40 \times 0.02}{0.0345}$$

$$= 0.6377 \approx 0.64$$

[Probability of 3 machines sum should be 1. Here, $1 - 0.36 = 0.64$]

Lecture Notes-1)

Example: 1.9: [Radar Detection] [Pg-21]

Check Ques from the Lecture Notes-1.pdf. Check the ans.

~~pg 31~~
 A : The aircraft is present; A^c : aircraft was not present
 B : The radar registers an aircraft presence

i) $P(A) = P(\bar{A}) = 0.05$ [$P(A_2) = 1 - P(A_1) = 0.95$]
or registered

ii) $P(B | A) = 0.99$ [radar detected given aircraft present]

iii) $P(B | A^c) = 0.10$ [$A_1, A_2 \dots$ different events. If that's why
 0.10 not 0.01 as not related events or probability]

iv) $P(\text{aircraft present} | \text{radar registers})$

$= P(A | B)$ corrected into.

$$\therefore P(A_1|\bar{B}) = \frac{P(A_1) \cdot P(B|A_1)}{P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2)}$$

$$= \frac{0.05 \times 0.99}{(0.05 \times 0.99) + (0.95 \times 0.1)}$$

$$= \frac{0.0495}{0.0495 + 0.095} = \frac{0.0495}{0.1445}$$

$$= 0.34256$$

$$P(A_2|\bar{B}) = P(\text{Aircraft absent} | \text{Radar registers})$$

$$= 0.66$$

False
Alarm

$$0.095 = \frac{P(A^c \cap B)}{\text{false alarm}} = \frac{P(A^c) \cdot P(B|A^c)}{P(B|A^c)}$$

$$\frac{0.0005}{0.005} = \frac{P(A \cap B^c)}{\text{Missed Detection}} = \frac{P(A) \cdot P(B^c|A)}{P(B^c|A)} \rightarrow 0.01$$